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LRSP-Fusion: A general image fusion network based on low-rank and sparse priors

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ABSTRACT

Multi-source images captured from the same scene typically contain both common and unique features, which has been widely exploited in specific image fusion tasks. Existing methods typically consider rough separation of common and unique features for a specific fusion task. However, due to the different modal characteristics of modal images in different image fusion tasks, existing feature separation methods are not suitable to general image fusion tasks. To address this issue, we propose an interpretable general image fusion network based on low rank and sparse priors (LRSP-Fusion) for multi-source image fusion tasks. Firstly, based on the principle that image matrices are usually low rank or approximately low rank, a common feature encoder (CFE) is designed based on low-rank matrix factorization theory to extract common features from multi-source images. Then, according to the sparsity of unique features, the optimization models based on sparse constraints are defined, and their corresponding unique feature encoders (UFEs) are designed by unfolding the iterative process of the sparse constraint problem into a neural network to learn the unique features of multi-source images. In addition, to achieve precise fusion of unique and common features, an entropy-based feature fusion block (EFFB) is constructed to achieve the fusion of features from CFE and two UFEs. A large number of experiments on different fusion tasks such as infrared and visible image fusion, medical image fusion, multi-exposure image fusion, and multi-focus image fusion show that our method outperforms some state-of-the-art methods in both objective and subjective aspects.

1. Introduction

Due to the different imaging mechanisms of various sensors, images captured from the same scene often contain complementary information (Yuan et al., 2025; Zhang et al., 2021b, 2024, 2020c). For example, infrared sensors highlight significant targets by capturing thermal radiation, while visible sensors generate images with rich texture details through reflected light (Li & Wu, 2018; Tang et al., 2023a; Wu et al., 2020). RGB-T fusion has also been applied in dual-spectrum UAV remote sensing and ground object understanding, further demonstrating its practicability for mission scenarios under complex environmental conditions (Shi et al., 2025; Zhang et al., 2026). Additionally, positron emission tomography (PET) can reveal the metabolic activity of tissues, while magnetic resonance imaging (MRI) provides high-resolution anatomical structural information (Li et al., 2023b; Wen et al., 2023). A single sensor, under different exposure settings, can only capture a portion of the information within its dynamic range (Qu et al., 2022; Ram Prabhakar et al., 2017). Moreover, cameras with different focal

lengths can clearly present objects within the depth of field (Zhang, 2021b). The goal of image fusion is to integrate the complementary information from multi-source images and generate a fused image that offers rich information. Image fusion tasks can be broadly divided into two major categories. The first is multi-modality image fusion (MMIF), which mainly includes infrared-visible image fusion (IVF) and medical image fusion (MIF). The second is digital photography image fusion (DPIF), which encompasses multi-exposure image fusion (MEF) and multi-focus image fusion (MFF). Researchers have designed various network architectures for specific tasks and achieved significant results (Huang et al., 2025; Li et al., 2020, 2021). However, networks designed for a specific fusion task often have weak generalization and are not suitable for other fusion tasks (Xu et al., 2023; Zhu et al., 2022).

The goal of different image fusion tasks is similar (Xu et al., 2020; Zhang & Ma, 2021; Zhang et al., 2020a), which is to generate fused images with rich information by fusing images of different modalities. Therefore, a general image fusion framework is of practical significance because it can reduce the need to separately design and

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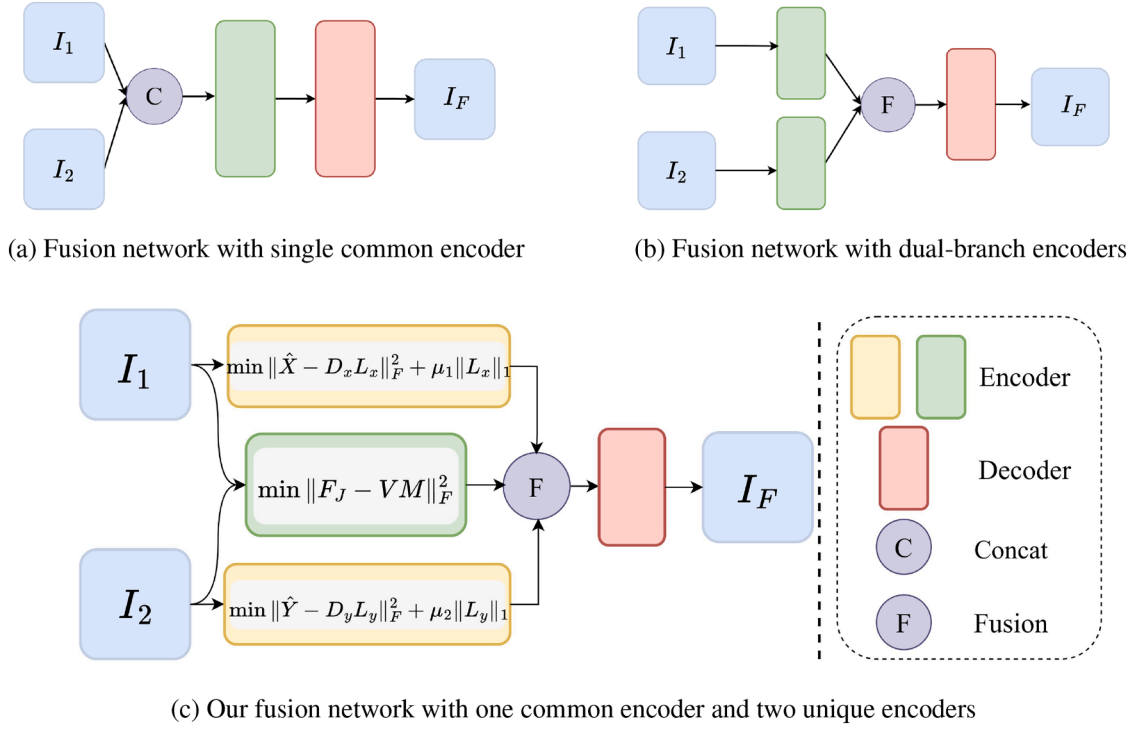


Fig. 1. Existing general image fusion methods vs. LRSP-Fusion.

maintain task-specific models for different fusion scenarios, while improving the adaptability of the model in practical multi-scenario vision systems. Many studies have focused on a universal method for unifying multi-source image fusion tasks (Cheng et al., 2023; Xing et al., 2024; Xu et al., 2022, Liu et al., 2024; Zhang et al., 2025). Existing general fusion methods suffer from the following limitations: 1). As shown in Fig. 1(a), a single common encoder is used to capture common features across modalities, which will overlook modality-specific characteristics. In contrast, the dual-encoder design in Fig. 1(b) emphasizes the modal characteristics of different features while ignoring the shared representation of features in the same scene. 2). From the perspective of network architecture, the internal structure often involves stacked Transformer (Li et al., 2024; Tang et al., 2023b) and CNN (Luo et al., 2023) operations, leading to excessive computational complexity and reduced fusion efficiency.

To extract more fine-grained features, task-specific methods have extensively explored the commonalities and differences between modalities, thus developing the feature decomposition-based approaches (Xu et al., 2021a; Zhao et al., 2023b, 2020). These methods typically decompose the input multimodal images into rough common and unique features (Deng & Dragotti, 2020; Li et al., 2023a). For example, LRRNet (Li et al., 2023a), designed for IVF, tackles feature separation from a mathematical perspective, modeling background information as a low-rank structure and salient information as a sparse structure. DeFusion (Liang et al., 2022) introduces a modality decomposition strategy that separates source images into shared components across modalities and modality-specific components, thereby offering a new perspective to general image fusion methods. However, DeFusion employs coarse binary masks to distinguish between common and unique features, which can lead to inaccurate decomposition. In image restoration, an image can be decomposed into a low-rank matrix representing the background and a sparse matrix representing abnormal regions (Wang et al., 2020). For image fusion, the source images are captured from the same scene and thus share identical background information, while exhibiting modality-specific characteristics. Therefore, the common features across source images can be represented as a shared low-rank matrix, whereas

the modality-specific features correspond to sparse matrices capturing modality-dependent variations.

Based on the above analysis, we extend the low-rank and sparse prior modeling to general image fusion tasks and propose a universal fusion framework, termed LRSP-Fusion, which solves the image fusion problem by exploring the intrinsic relationships of source image features. Compared with previous methods, our approach provides two key advantages. First, as shown in Fig. 1(c), from a framework perspective, our method not only enables effective cross-modal interaction but also fully exploits modality-specific features. Second, from a feature decomposition perspective, we achieve feature disentanglement through rigorous mathematical formulations, thereby avoiding reliance on empirically designed network structures. To be specific, LRSP-Fusion consists of a two-stage encoder and a two-stage decoder. In the encoder, a common feature encoding stage employs a low-rank decomposition based common feature encoder (CFE) to extract common features, while a unique feature encoding stage extracts the unique features of two source images by constructing two sparse representation-based unique encoders (UFEs). In the decoder, a feature fusion stage achieves the fusion of unique and common features of the source images by constructing an entropy-based feature fusion block (EFFB), and a feature reconstruction stage is constructed to obtain the fused image. Our contributions are summarized as follows.

- A general image fusion network based on low-rank and sparse priors, named LRSP-Fusion, is proposed to solve the common problems in fusion tasks by modeling the common and unique features as low rank approximation and sparse constraint problems.
- A CFE is constructed by utilizing the Non-negative Matrix Factorization (NMF) to extract common features of the input source images, and a UFE is designed by unfolding the iterative process of the sparse constraint problem into a neural network to extract unique features of different source image. An EFFB is constructed to adaptively fuse unique features by calculating fusion coefficients using the entropy of different modalities.

- Experimental results on multiple public benchmark datasets show our method outperforms multiple state-of-the-art (SOTA) methods, including both general and task-specific approaches.

2. Related works

2.1. Feature decomposition-based image fusion methods.

In image fusion tasks, multi-source images from the same scene typically contain common features (modality redundancy) and unique features (modality complementarity). Leveraging this inherent property, numerous studies have proposed feature decomposition methods that significantly enhance fusion performance. These methods can generally be categorized into two main types. One type is the loss constraint-based methods. DRF (Xu et al., 2021b) employs a subtraction-based loss function to minimize the distance between common features (scene information) while increasing the separation between unique features and between common and unique features, thereby achieving more explicit feature disentanglement. CDDFuse (Zhao et al., 2023b) defines common features as low-frequency background information and unique features as high-frequency gradient details, introducing a division-based decoupling loss to effectively separate them. Conti-Fuse (Li et al., 2025) performs repeated decomposition in the feature space to obtain more diverse and enriched feature representations. However, these methods are often task-specific and lack generality, limiting their applicability to a wide range of image fusion tasks. The other type is the network structure-based methods. DeFusion (Liang et al., 2022) proposes a general image fusion framework by applying random masks to the input images, treating non-overlapping regions as unique features and overlapping regions as common features. It utilizes two independent encoders to extract unique features and a shared encoder to project common regions into a unified feature space. However, this mask-based coarse decomposition may result in the loss of critical information, adversely affecting the fusion performance. The above decomposition methods largely rely on heuristic designs. LRRNet (Li et al., 2023a) addresses this limitation through mathematical modeling, representing background and salient information as low-rank and sparse components, respectively. It unfolds the low-rank and sparse dictionary learning process into a CNN-based framework, offering an efficient and interpretable architecture for IVF.

2.2. General image fusion methods.

Since different image fusion tasks share a common objective, several general methods have been proposed to improve network adaptability. Architecturally, these methods can be classified into two categories: CNN-based methods and Transformer-based methods. PMGI (Zhang et al., 2020a) employs a dual-branch feature extraction network to capture modality-specific information and formulates multi-source image fusion as a task of preserving both pixel intensities and gradient details. It manually assigns weight ratios to different tasks. To overcome the limitations of fixed ratios, U2Fusion (Xu et al., 2022) utilizes a pre-trained VGG network to extract deep features from source and fused images, incorporating image quality assessment to evaluate the importance of each feature, thus enabling adaptive weighting. Although CNNs are effective at extracting local features, they are limited in capturing global context. SwinFusion (Ma et al., 2022) addresses this by leveraging self-attention and cross-attention mechanisms to facilitate both intra- and cross-modality global information interaction, thereby improving fusion performance. TUFusion (Zhao et al., 2023a) integrates Transformer and CNN architectures to build a hybrid encoder that captures global representations while preserving local details, further enhancing fusion quality.

3. The proposed algorithm

3.1. Motivation

As previously discussed, multi-source images captured from the same scene typically contain both common and unique features. This property provides a solid foundation for developing general image fusion methods. In task-specific approaches, extensive studies have shown that common features generally correspond to background information and are primarily composed of low-frequency components, whereas unique features are characterized by salient textures in the high-frequency domain. LRRNet further models this observation from a mathematical perspective, defining background information as a low-rank structure and salient information as a sparse structure. Inspired by this, we define the common features of multi-source images as low-rank background components, and the unique features as sparse elements that deviate from the background.

However, unlike the decomposition strategy adopted in LRRNet, we propose a novel iterative solution approach and develop a general image fusion framework. Given a pair of multi-source images, we first project them into a unified feature space F_j . To extract low-rank common features, we apply NMF to approximate F_j , aiming to capture the global low-rank structure that represents common information across modalities. Specifically, NMF decomposes F_j into the product of a dictionary matrix V and a coefficient matrix M , such that $F_j \approx VM$, effectively capturing the latent low-rank features that represent common features. To extract unique features, we first compress the common representation and align it with the input images. Then, by performing element-wise subtraction—removing background content from the inputs—we obtain an initial estimate of the unique components. Given that CNN-based models often suffer from limited interpretability and inconsistent feature extraction across modalities (Zhao et al., 2023b), we explicitly unfold the unique feature extraction process into a neural network architecture to enhance transparency and control. Building on this strategy, we propose a lightweight, general and interpretable image fusion method.

3.2. Structure of LRSP-Fusion

In this section, we propose a general image fusion network called LRSP-Fusion, as shown in Fig. 2, which consists of a two-stage encoder and a two-stage decoder. In the encoder, considering that the two source images have the same background features and unique modal features, a low-rank decomposition-based CFE is constructed to extract common features Φ_S of two images X and Y , while two sparse representation-based UFEs are designed to extract unique features Φ_U^X and Φ_U^Y of the two images separately. The common features are compressed through a 1×1 convolution to obtain a common feature map, which is then subtracted from the input source image to obtain the initial unique features. The specific operation of the encoder can be represented by the following equations:

$$\Phi_S = CEn(Cat(X, Y)) \quad (1)$$

$$\Phi_U^X = UEn(X - Conv_{1 \times 1}(\Phi_S)) \quad (2)$$

$$\Phi_U^Y = UEn(Y - Conv_{1 \times 1}(\Phi_S)) \quad (3)$$

where $CEn(\cdot)$ and $UEn(\cdot)$ denotes the operations of CFE and UFE, respectively. $Cat(\cdot)$ and $Conv_{1 \times 1}(\cdot)$ denote the concatenation and 1×1 convolution operations.

In the decoder, firstly, a SAFM containing two entropy-based feature fusion blocks (EFFBs) is constructed to sequentially achieve the fusion of unique features of two images, as well as the fusion of fused

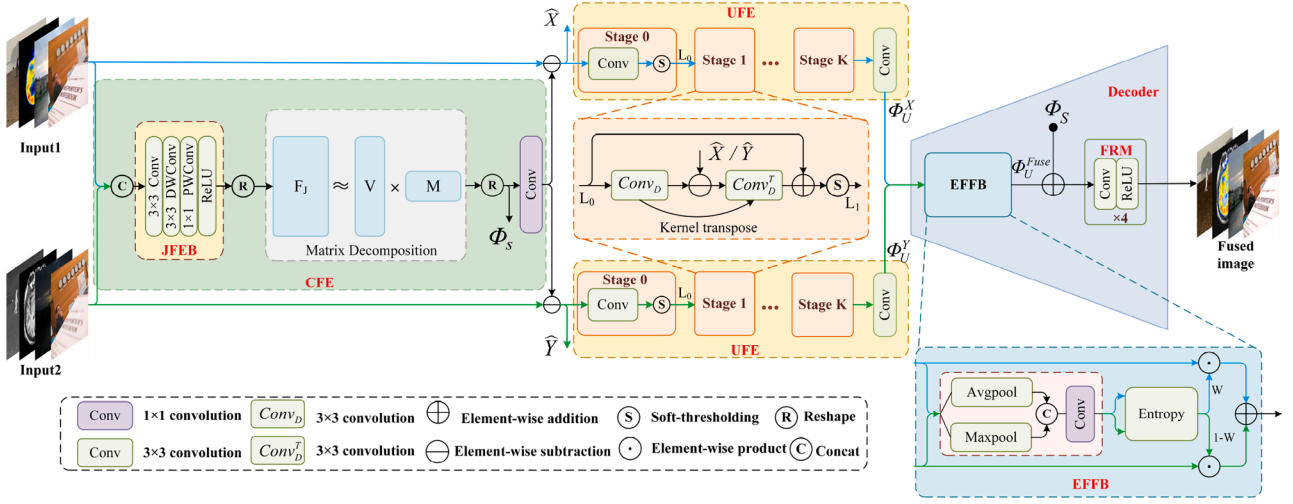


Fig. 2. The architecture of proposed LRSP-Fusion.

unique features and common features. Then, the fused features F_f are fed into a feature reconstruction module (FRM) containing multiple convolutional layers to obtain the final fusion result I_f . Each convolutional layer $ConL(\cdot)$ contains a 3×3 convolution and a ReLU function. The operations of the decoder can be represented as follows:

$$F_f = EFFB(EFFB(Cat(\Phi_U^X, \Phi_U^Y)), \Phi_S) \quad (4)$$

$$I_f = FRM(F_f) = ConL(\dots(ConL(F_f))) \quad (5)$$

where $EFFB(\cdot)$ and $FRM(\cdot)$ denote the operations of EFFB and FRM, respectively. Below, we provide a detailed introduction to the main components in LRSP-Fusion, such as CFE, UFE, and EFFB.

3.3. Common feature encoder (CFE)

In low rank matrix recovery, the data matrix of a natural image is often low rank or approximately low rank, and the low-rank matrix represents the main structure and information of the image. In image fusion tasks, the input source images are all obtained from the same scene, and they have the same basic structure and information, which can be regarded as common features of the source images and have the same low-rank representation. In addition, NMF is a low-rank approximation model that can extract potential features of data, and image matrix is also a non-negative matrix. Therefore, this paper adopts NMF technology to achieve low-rank decomposition of image matrix, and utilizes the decomposed V (basis matrix) and M (coefficient matrix) to reconstruct common features. Based on above idea, a low-rank decomposition-based CFE is constructed to extract common features of the input source images. The execution process of CFE is as follows. Firstly, the input source images are sent together to a joint feature extraction block (JFEB) to extract the shallow joint features F_J . JFEB is composed of a 3×3 convolution and a depthwise separable convolution consisting of depthwise convolution (DWConv) and pointwise convolution (PWConv), which can reduce the number of model parameters and computational complexity.

$$F_J = PWConv_{1 \times 1}(DWConv_{3 \times 3}(Con_{3 \times 3}(Cat(X, Y)))) \quad (6)$$

Then, NMF technology is employed to obtain the low-rank approximation of the joint features F_J , namely $V \in \mathbb{R}^{HW \times R}$ and $M \in \mathbb{R}^{R \times C}$, where $R < \min(HW, C)$. Specifically, NMF technique approximates the non-negative feature matrices F_J by minimizing the reconstruction error. The common loss function is mean square error, which is defined as follows:

$$\min_{V, M \geq 0} \|F_J - VM\|_F^2 \quad (7)$$

where $\|\cdot\|_F$ denotes the Frobenius norm of the matrix, with the constraint that all elements of V and M are greater than or equal to 0. Mechanistically, the low-rank constraint is introduced to encourage a compact representation of the dominant correlated structure shared by both modalities. Since the two source images are captured from the same scene, their stable background layout and major structural contours are more likely to be repeatedly represented in the shared low-rank subspace, whereas modality-specific deviations are less likely to be absorbed by the compact factors. The multiplication update rule is a commonly used NMF optimization method that minimizes the loss function by iteratively updating V and M (Geng et al., 2021). The specific update formulas are as follows:

$$V^k = V^{k-1} \odot \frac{F_J(M^{k-1})^T}{V^{k-1}M^{k-1}(M^{k-1})^T} \quad (8)$$

$$M^k = M^{k-1} \odot \frac{(V^k)^T F_J}{(V^k)^T V^k M^{k-1}} \quad (9)$$

where $\odot$ denotes Hadamard product, k denotes the k -th iteration, and T denotes the matrix transposition operation. Finally, by multiplying V and M , common features Φ_S are obtained, which can be expressed as follows:

$$\Phi_S = VM \quad (10)$$

3.4. Unique feature encoder (UFE)

UFE aims to extract refined unique features. The coarse unique features of each source image are obtained by subtracting the integrated common features from the source images. Here, a 1×1 convolution is used to integrate common features through channel dimensionality reduction.

$$\hat{X} = X - Conv_{1 \times 1}(\Phi_S), \hat{Y} = Y - Conv_{1 \times 1}(\Phi_S) \quad (11)$$

where $\hat{X}$ and $\hat{Y}$ denote the coarse unique features of source images X and Y , respectively. Common features are the main structure and information of an image, while unique features represent the modal information of the source image, which is sparse. Therefore, UFE is constructed based on sparse representation of unique features to solve the following sparse decomposition model:

$$\begin{aligned} \min_{D_x, L_x} & \|\hat{X} - D_x L_x\|_F^2 + \mu_1 \|L_x\|_1 \\ \min_{D_y, L_y} & \|\hat{Y} - D_y L_y\|_F^2 + \mu_2 \|L_y\|_1 \end{aligned} \quad (12)$$

where $\|\cdot\|_1$ denotes the ℓ_1 -norm, L_x , and L_y represent the sparse coefficients of $\{\hat{X}, \hat{Y}\}$, respectively, and D_x , and D_y denote the corresponding

dictionary. Eq. (12) can be solved using the traditional method in Gregor and LeCun (2010). Here, subtracting the compressed common features from each source image provides a coarse modality-specific residual, and the subsequent sparse constraint further encourages the model to preserve only salient deviations from the shared background. In this way, UFE gradually suppresses redundant common responses while retaining the informative modality-dependent details. The iterative solution formulas are shown below:

$$L_x^k = S_{\mu_1}(L_x^{k-1} + D_x^T(\hat{X} - D_x L_x^{k-1})) \quad (13)$$

$$L_y^k = S_{\mu_2}(L_y^{k-1} + D_y^T(\hat{Y} - D_y L_y^{k-1})) \quad (14)$$

where $S_{\mu}(\cdot)$ denotes the soft-thresholding operation, defined as follows:

$$S_{\mu}(x) = \text{sign}(x) \cdot \max(|x| - \mu, 0) \quad (15)$$

where $\text{sign}(\cdot)$ denotes the sign function and $\max(\cdot)$ denotes the function for finding the maximum value, and μ is the threshold to shrink the input x .

Inspired by Sreter and Giryas (2018), the matrix multiplication in Eqs. (13) and (14) can be replaced by convolutional operations. Consequently, these equations can be unfolded into a neural network with K stages, where D_x and D_x^T are replaced by convolutional layers Conv_D and Conv_D^T , respectively. Similarly, D_y and D_y^T follow the same transformation. As shown in Fig. 2, each iteration ($k = 1, \dots, K$) corresponds to one step in the solving process, which can be represented as a kernel transposition convolution block (KTCB) and is formulated as follows:

$$L_x^k = S_{\mu_1}(L_x^{k-1} + \text{Conv}_D^T * (\hat{X} - \text{Conv}_D * L_x^{k-1})) \quad (16)$$

$$L_y^k = S_{\mu_2}(L_y^{k-1} + \text{Conv}_D^T * (\hat{Y} - \text{Conv}_D * L_y^{k-1})) \quad (17)$$

where μ_1 and μ_2 are the learnable parameters, $*$ denotes the convolution operation, Conv_D denotes 3×3 convolution, and Conv_D^T denotes its kernel-transposed version. Here, stage 0 is constructed to initialize L_x^0 and L_y^0 , as shown below:

$$L_x^0 = S_{\mu_1}(\text{Conv}_{3 \times 3} * \hat{X}), L_y^0 = S_{\mu_2}(\text{Conv}_{3 \times 3} * \hat{Y}) \quad (18)$$

The sparse coefficients L_x^K and L_y^K from stage K in two UFEs are processed separately through a 3×3 convolution to obtain sparse unique features Φ_U^X and Φ_U^Y , which are represented as:

$$\Phi_U^X = \text{Conv}_{3 \times 3} * L_x^K, \Phi_U^Y = \text{Conv}_{3 \times 3} * L_y^K \quad (19)$$

The solution of common and unique features in the source images can be represented by Algorithm 1.

Algorithm 1 The solution of common and unique features in LRSP-Fusion.

Require: Source images $\{X, Y\}$

- 1: **Generate common features** Φ_S
- 2: (a) Obtain joint features F_J using JFEB, as shown in Eq. (6)
- 3: (b) Compute basis matrix V and coefficient matrix M via NMF Eq. (8) and Eq. (9)
- 4: (c) Obtain common features Φ_S from Eq. (10)
- 5: **Generate unique features** Φ_U^X and Φ_U^Y
- 6: Initialize coarse unique features with Eq. (11), and sparse coefficients L_x^0 and L_y^0 with Eq. (18)
- 7: **for** $k = 1$ to K **do**
- 8: Update L_x and L_y using Eq. (16) and Eq. (17)
- 9: **end for**
- 10: **Output** Φ_U^X, Φ_U^Y using Eq. (19)

3.5. Entropy-based feature fusion block (EFFB)

An EFFB is constructed to achieve the fusion of unique features Φ_U^X and Φ_U^Y of two images. The entropy of an image reflects the average amount of information in the image, and the higher the entropy value

of a feature map, the richer the features it contains. Therefore, we use entropy calculation to evaluate the amount of information in the feature maps and obtain adaptive weights for the two unique features. Firstly, global average pooling and max pooling are used to obtain the details and bright-ness information of two sets of feature maps, and then 1×1 convolutions are used to obtain the integrated features F_X and F_Y . The above operations can be represented as follows:

$$F_X = \text{Conv}_{1 \times 1}(\text{Ap}(\Phi_U^X), \text{Mp}(\Phi_U^X)) \quad (20)$$

$$F_Y = \text{Conv}_{1 \times 1}(\text{Ap}(\Phi_U^Y), \text{Mp}(\Phi_U^Y)) \quad (21)$$

where $\text{Ap}(\cdot)$ and $\text{Mp}(\cdot)$ denote the operations of global average pooling and max pooling. Next, the entropies of the integrated features F_X and F_Y are calculated and used to calculate the fusion weights w_a and w_b of the two features using the following formulas:

$$\{w_a, w_b\} = \left\{ \frac{\text{En}(F_X)}{\text{En}(F_X) + \text{En}(F_Y)}, \frac{\text{En}(F_Y)}{\text{En}(F_X) + \text{En}(F_Y)} \right\} \quad (22)$$

where $\text{En}(\cdot)$ denotes the operation for calculating entropy. Finally, the two unique features are weighted using fusion weights and then added together to obtain the fused unique feature Φ_U^{fuse} , and the operation is defined as follows:

$$\Phi_U^{\text{fuse}} = w_a \Phi_U^X + w_b \Phi_U^Y \quad (23)$$

3.6. Loss function

To better guide network training, a joint loss function consisting of content loss $\mathcal{L}_{\text{pixel}}$, gradient loss $\mathcal{L}_{\text{grad}}$, and structure loss $\mathcal{L}_{\text{ssim}}$ is defined as follows:

$$\mathcal{L}_{\text{total}} = \alpha_1 \mathcal{L}_{\text{pixel}} + \alpha_2 \mathcal{L}_{\text{grad}} + \alpha_3 \mathcal{L}_{\text{ssim}} \quad (24)$$

where $\{\alpha_1, \alpha_2, \alpha_3\}$ are the tuning parameters. The definition of content loss $\mathcal{L}_{\text{pixel}}$, gradient loss $\mathcal{L}_{\text{grad}}$, and structure loss $\mathcal{L}_{\text{ssim}}$ are as follows:

$$\mathcal{L}_{\text{pixel}} = \frac{1}{HW} \|I_f - AG(X, Y)\|_1 \quad (25)$$

$$\mathcal{L}_{\text{grad}} = \frac{1}{HW} \|\nabla|I_f| - \max(\nabla|X|, \nabla|Y|)\|_1 \quad (26)$$

$$\mathcal{L}_{\text{ssim}} = \frac{1}{2} [(1 - SSIM(I_f, X)) + (1 - SSIM(I_f, Y))] \quad (27)$$

where ∇ indicates the Sobel gradient operator, and $AG(\cdot)$ is an element-wise aggregation operation. For IVF and MIF tasks, $AG(\cdot)$ is set as the operation to select the maximum value, i.e. $\max(\cdot)$. For MEF and MFF tasks, $AG(\cdot)$ is set as the operation for calculating the mean, i.e., $\text{mean}(\cdot)$.

4. Experiments

Our experiments are conducted on a machine with an NVIDIA RTX A6000 GPU. We train the model for 300 epochs, with an initial learning rate of 2×10^{-4} and decreasing by a factor of 0.1 every 100 epochs. The Adam optimizer is used, with a batch size set to 32. For network hyperparameters setting, the number of feature channels C is set to 32. In CFE, the number of iterations for matrix decomposition is set to 6, and the rank R is set to 8.

Our experiments are conducted on a machine with an NVIDIA RTX A6000 GPU. The proposed model is trained for 100 epochs with an initial learning rate of 0.001, using the Adam optimizer and a batch size of 32. For network hyperparameters, $\{\alpha_1, \alpha_2, \alpha_3\}$ in Eq. (24) are set to 20, 20, 10, the number of iterations S for matrix decomposition in CFE is set to 6, with the rank R fixed at 16. In UFE, the number of iteration stage K is set to 3 for the IVF task, and to 2 for the MEF and MFF tasks. We evaluate LRSP-Fusion on two fusion scenarios: MMIF and DPIF. For MMIF, the model is trained on the IVF task and directly tested on the medical image fusion (MIF) task without fine-tuning. The IVF training set consists of 1083 image pairs from the MSRS (Tang et al., 2022a) dataset. The test sets include 42 pairs from TNO, 20 pairs from Road-Scene (Xu et al., 2022), and 50 pairs from M3FD (Liu et al., 2022).

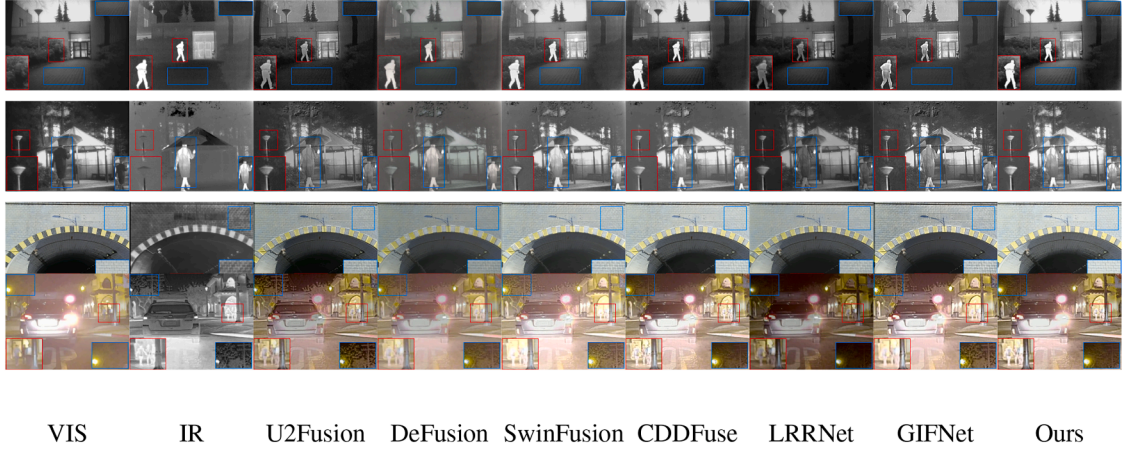


Fig. 3. Qualitative comparison of our method with SOTA fusion methods on the TNO, M3FD, and RoadScene datasets in the IVF task.

Table 1

Comparison of quantitative results in the IVF task on three datasets.

Dataset	Metric	U2Fusion	DeFusion	SwinFusion	CDDFuse	LRRNet	GIFNet	Ours
TNO	NMI	0.299	0.392	0.504	0.469	0.375	0.294	0.587
	SD	37.697	30.502	39.648	<u>44.529</u>	40.879	40.409	48.026
	MI	2.005	2.633	<u>3.377</u>	3.127	2.517	1.972	3.948
	VIF	0.677	0.660	0.837	<u>0.842</u>	0.623	0.588	0.861
	EN	6.997	6.580	6.907	<u>7.069</u>	6.988	6.920	7.169
M3FD	NMI	0.401	0.451	0.565	0.626	0.460	0.401	0.674
	SD	34.825	29.365	38.823	<u>41.169</u>	30.774	40.773	44.102
	MI	2.760	3.095	3.884	<u>4.310</u>	3.167	2.753	4.637
	VIF	0.761	0.633	0.733	0.844	0.651	0.639	<u>0.809</u>
	EN	6.894	6.514	6.911	<u>7.031</u>	6.594	6.979	7.038
RoadScene	NMI	0.393	0.466	0.492	0.452	0.369	0.405	0.529
	SD	44.972	41.434	53.307	<u>56.694</u>	41.237	55.161	56.752
	MI	2.812	3.327	<u>3.499</u>	3.224	2.642	2.899	3.769
	VIF	0.684	0.713	0.784	0.730	0.535	0.619	<u>0.764</u>
	EN	7.362	7.175	7.338	7.541	7.101	7.560	<u>7.558</u>

For MIF, 27 image pairs of MRI–PET and MRI–SPECT are used for testing, obtained from the Harvard Medical website. For DPIF, we conduct experiments on both multi-exposure fusion (MEF) and multi-focus fusion (MFF). In MEF, the model is trained on 340 image pairs from the SICE (Cai et al., 2018) dataset and tested on 40 pairs from SICE and 50 pairs from the MEFB (Zhang, 2021a) dataset. In MFF, training is performed on 800 image pairs from the MFI-WHU (Zhang et al., 2021a) dataset, and testing is conducted on 50 pairs from the RealMFF (Zhang et al., 2020b) dataset. Five objective evaluation metrics are employed to assess the fusion performance: normalized mutual information (NMI), standard deviation (SD), mutual information (MI), visual information fidelity (VIF), and entropy (EN). We compare our method with five representative SOTA general image fusion methods: U2Fusion (Xu et al., 2022), DeFusion (Liang et al., 2022), SwinFusion (Ma et al., 2022), TU-Fusion (Zhao et al., 2023a), and GIFNet (Cheng et al., 2025). Additionally, for MMIF, we compare our method with one SOTA MMIF method, CDDFuse (Zhao et al., 2023b), and one task-specific method, LRRNet (Li et al., 2023a). For MEF, we compare our method with HoLoCo (Liu et al., 2023), and for MFF, we compare it with YDTR (Tang et al., 2022b).

4.1. Results on multi-modal image fusion

1) Qualitative comparison. We present the qualitative results of the IVF task on three test datasets in Fig. 3. As shown in the figure, our method demonstrates clear advantages in preserving the salient information of infrared images while simultaneously retaining the fine-grained texture details of visible images. This can be attributed to the effective disentanglement and exploitation of both common and

modality-unique features within our framework. In contrast, due to the limited feature extraction capacity of conventional CNN architectures, U2Fusion fails to adequately highlight salient infrared targets and produces results with artifacts. Although SwinFusion and GIFNet benefit from the global modeling capability of Transformer architectures, they introduce additional noise in certain regions, as shown in the enlarged areas within the blue boxes in the third row of Fig. 3. DeFusion relies on coarse binary masks to separate common and unique features, but the fusion results suffer from low contrast and loss of texture information, which also highlights the limitations of its feature decomposition strategy. In addition, CDDFuse, which defines common features as low-frequency background components and unique features as high-frequency details, performs well in highlighting salient information, but sacrifices background texture fidelity. LRRNet only uses CNN for feature extraction, showing limited ability in emphasizing infrared salient regions, and has the problem of losing background details. Similarly, GIFNet also suffers from the loss of significant information and background texture information. Compared with these SOTA methods, our approach achieves a better balance between preserving salient infrared targets and maintaining visible texture richness, thereby producing visually more faithful and informative fusion results. In addition, to further validate the generalization of the model, we directly apply the trained model to the MIF task. Fig. 4 shows the qualitative comparison results of various comparison methods on the Harvard Medical dataset in the MIF task. It can be observed that even without task-specific fine-tuning, our network trained solely on the IVF task still produces fusion results that are comparable to those achieved by other SOTA methods.

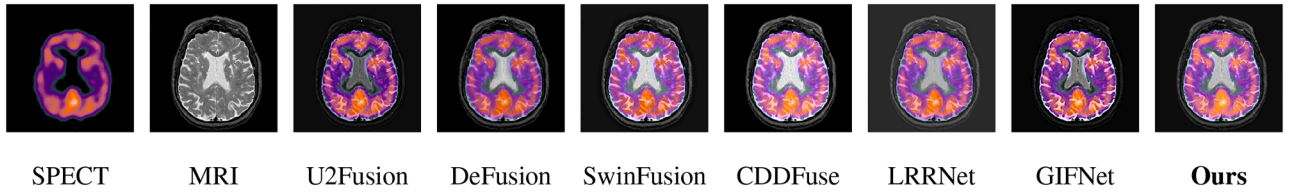


Fig. 4. Qualitative comparison of our method with SOTA fusion methods on the Harvard Medical dataset in the MIF task.

Table 2

Comparison of quantitative results for the MIF task on the Harvard Medical dataset.

Method	U2Fusion	DeFusion	SwinFusion	CDDFuse	LRRNet	GIFNet	Ours
NMI	0.753	0.817	0.821	0.886	0.772	0.737	0.895
SD	66.972	74.656	87.087	<u>85.691</u>	57.079	79.719	81.884
MI	2.712	2.943	2.956	<u>3.199</u>	2.781	2.655	3.233
VIF	0.465	0.59	0.618	<u>0.634</u>	0.508	0.404	0.64
EN	5.544	5.405	<u>5.729</u>	5.472	5.549	5.514	5.92

2) Quantitative comparison: Table 1 presents the quantitative evaluation results of our method compared with six other SOTA approaches on three benchmark datasets. On the TNO dataset, our method consistently achieves the best performance across all evaluation metrics, clearly demonstrating its superiority. On the M3FD dataset, our method ranks first on four out of five metrics, with only a slight drop to second place in terms of IVF, which highlights its overall robustness and competitiveness. On the RoadScene dataset, our method obtains three first ranked indicators and two second ranked indicators. Compared with the best performing method, the marginal difference in indicator values indicates that our method has strong generalization ability in different scenarios. In terms of metric-specific analysis, the highest NMI and MI scores demonstrate that our method effectively captures and integrates complementary information from both modalities. Moreover, the highest SD values further highlight the capability of our method to preserve fine-grained structural and textural details. Meanwhile, the strong performance on VIF and EN indicates that our fused images not only retain rich visual content but also align more closely with the human visual perception system, thereby producing results that are both informative and visually natural. Table 2 presents the quantitative results on the Harvard Medical dataset in the MIF task. Our method demonstrates superior performance, obtaining four first ranked indicators and one second ranked indicator among the five evaluation indicators.

4.2. Results on multi-exposure image fusion

(1) Qualitative comparison: Fig. 5 presents the qualitative comparison results for the MEF task. It can be observed that U2Fusion, DeFusion, and TUFusion suffer from local overexposure or underexposure, which leads to the loss of image details. SwinFusion performs well on the MEFB dataset, as shown in the first row. However, this method exhibits noticeable overexposure or underexposure on the SICE dataset, as shown in the second to fourth rows, indicating weak generalization of the method. The fusion results of GIFNet exhibit significant color distortion, further highlighting the limitations of this method when applied to MEF tasks. HoLoCo, specifically designed for the MEF task, produces generally satisfactory results compared to other comparison methods, but still has the issue of low contrast, as shown in the result in the fourth row. In contrast, our method demonstrates significant advantages in both color fidelity and preservation of local details. As illustrated in the red box in the fourth row, our approach preserves the finest textural details more effectively than competing methods, ensuring a balanced exposure and rich color representation of the entire fused image. These results on two datasets demonstrate the robustness of our method in maintaining structural integrity and improving visual quality in multi-exposure fusion scenarios.

(2) Quantitative comparison: As shown in Table 3, our method demonstrates consistently strong performance across both MEF test datasets. Specifically, our method obtains three first ranked metrics and one second ranked metric on both datasets. Notably, the highest NMI, EN, and MI values indicate that our fusion results effectively retain the maximum amount of information from the source modality images and contain rich structural and textural details. These results underscore the capability of our method to integrate complementary information while preserving fine-grained details, which is critical for generating visually faithful and informative fused images. Moreover, the consistent ranking across different datasets highlights the robustness and generalization ability of our approach in multi-exposure image fusion scenarios.

4.3. Results on multi-focus image fusion

(1) Qualitative comparison: Fig. 6 illustrates the fusion results of three pairs of FAR and NEAR images from the RealMFF dataset. Our method successfully extracts reliable scene information from both focused and defocused regions, producing all-in-focus images with high contrast, sharp edges, and rich structural details. Especially, our approach maintains the overall chromatic distribution of the original images, ensuring that the fused results remain visually natural and consistent with the source modality information. In contrast, DeFusion and TUFusion fail to sufficiently highlight the sharp texture details, leading to blurred or less distinct boundaries. Although GIFNet generates sharper fused images, its results often deviate from the domains of the original modalities, such as excessive enhancement and distorted luminance distribution. For instance, in the third row, the ground area in the fused image appears brighter than the focused area in the source image, indicating the presence of brightness distortion in the fused result.

(2) Quantitative comparison: The quantitative evaluation results are summarized in Table 4. Our method achieves the best performance on four out of the five evaluation metrics, with only the SD value ranking second, slightly behind GIFNet. Nevertheless, the superior VIF score obtained by our method highlights that our fused images are more consistent with human visual perception. This observation aligns well with the qualitative results, indicating that our method consistently produces visually pleasing and perceptually faithful fusion outcomes.

4.4. Downstream applications

To further investigate the practical value of image fusion in high-level vision applications, we evaluate the performance of object detection task on infrared, visible, and corresponding fused images obtained by SOTA methods. Specifically, we employ YOLOv5 as the detector and fine-tune it on the visible images of the M3FD dataset. Representative



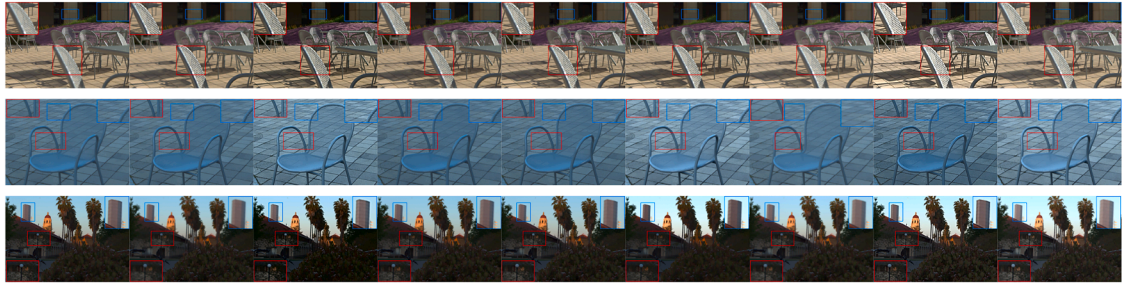
UNDER OVER U2Fusion DeFusion SwinFusion HoLoCo TUFusion GIFNet **Ours**

Fig. 5. Qualitative comparison of our method with SOTA fusion methods on the MEFB and SICE datasets in the MEF task.

Table 3

Comparison of quantitative results in the MEF task on the MEFB and SICE datasets.

Dataset	Method	U2Fusion	DeFusion	SwinFusion	HoLoCo	TUFusion	GIFNet	Ours
MEFB	NMI	0.997	0.898	0.897	0.748	<u>1.024</u>	0.99	1.028
	SD	54.343	45.624	46.442	48.553	49.581	70.773	<u>54.52</u>
	MI	6.027	5.492	5.438	4.525	<u>6.201</u>	5.997	6.223
	VIF	<u>1.369</u>	1.201	1.461	1.266	1.046	1.214	1.23
	EN	<u>7.185</u>	6.724	7.072	7.134	7.015	7.174	7.233
SICE	NMI	0.979	0.753	<u>0.989</u>	0.754	0.935	0.983	1.008
	SD	48.896	39.821	<u>40.865</u>	39.732	44.124	64.32	<u>50.018</u>
	MI	5.184	4.028	<u>5.245</u>	4.011	4.96	5.202	5.33
	VIF	1.257	0.975	<u>1.188</u>	0.979	0.758	1.087	1.072
	EN	6.906	6.648	6.294	6.855	6.818	<u>7.06</u>	7.068



FAR NEAR U2Fusion DeFusion SwinFusion YDTR TUFusion GIFNet **Ours**

Fig. 6. Qualitative comparison of our method with SOTA fusion methods on the RealMFF dataset in the MFF task.

Table 4

Comparison of quantitative results in the MFF task on REALMFF dataset.

Method	U2Fusion	DeFusion	SwinFusion	YDTR	TUFusion	GIFNet	Ours
NMI	0.881	<u>1.046</u>	1.042	0.915	0.993	0.819	1.048
SD	56.136	49.483	50.988	55.033	52.726	61.981	<u>56.227</u>
MI	6.196	<u>7.353</u>	7.327	6.44	6.985	5.76	7.365
VIF	1.352	1.399	<u>1.47</u>	1.233	1.271	1.147	1.485
EN	7.144	7.030	7.083	7.057	7.094	<u>7.205</u>	7.230

detection results on two sample images from the M3FD dataset are illustrated in Fig. 7. As highlighted in the enlarged regions, our method, similar to other SOTA approaches, significantly enhances the detection accuracy of various objects. In our results, the detection accuracy of mo-

st targets is higher than that of other results. This demonstrates that infrared and visible image fusion not only provides visually informative representations but also contributes to more robust and reliable performance in downstream computer vision tasks such as object detection.

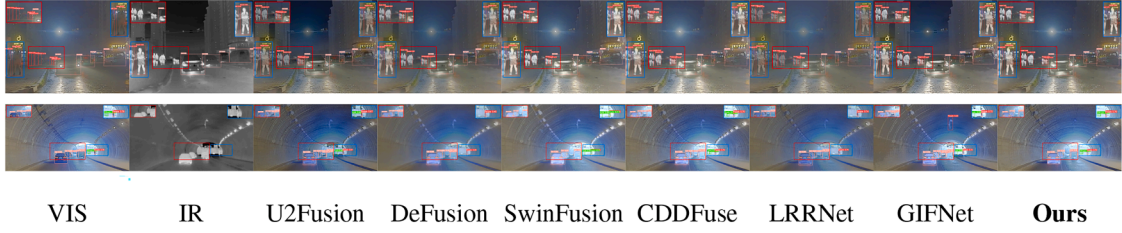


Fig. 7. Comparison of the object detection results of our method and six fusion methods on #00465 and #02651 images from the M3FD dataset.

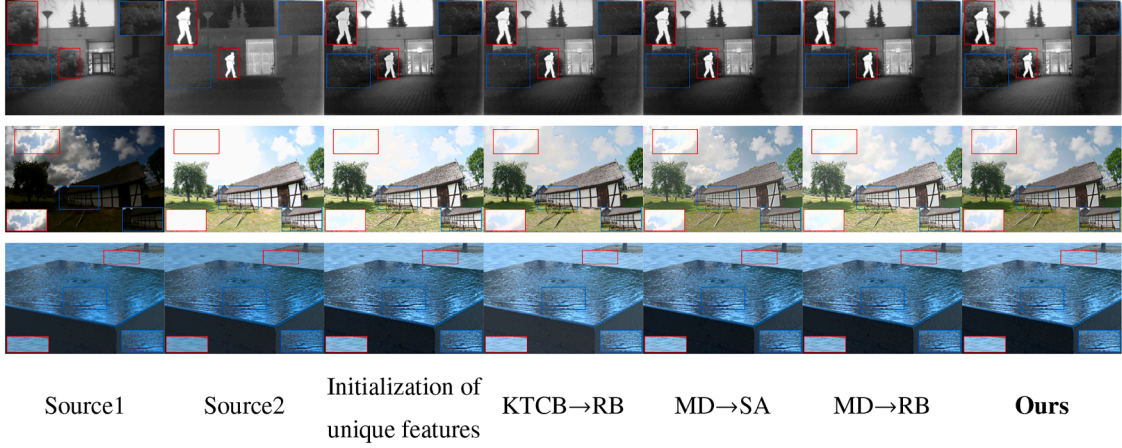


Fig. 8. Qualitative comparison of the ablation experiment on the IVF, MEF, and MFF tasks.

Table 5

Quantitative comparison of ablation studies on the IVF, MEF, and MFF tasks.

Task	Method	Initialization of unique features	KTCB → RB	MD → SA	MD → RB	Ours
IVF	NMI	0.577	0.549	0.554	<u>0.580</u>	0.587
	SD	50.605	41.738	43.047	42.489	<u>48.026</u>
	MI	3.878	3.702	3.736	<u>3.918</u>	3.948
	VIF	0.848	0.857	0.849	0.867	<u>0.861</u>
	EN	7.062	7.042	7.049	<u>7.065</u>	7.169
MEF	NMI	0.873	<u>0.937</u>	0.879	0.800	1.028
	SD	61.129	54.164	48.203	52.890	<u>54.520</u>
	MI	5.308	<u>5.707</u>	5.327	4.880	6.223
	VIF	1.147	<u>1.152</u>	<u>1.176</u>	1.071	1.230
	EN	6.505	6.909	<u>7.068</u>	6.930	7.233
MFF	NMI	1.034	1.040	1.041	<u>1.041</u>	1.048
	SD	53.939	50.605	53.945	53.029	56.227
	MI	7.275	<u>7.319</u>	7.316	7.317	7.365
	VIF	1.449	1.435	1.452	<u>1.449</u>	1.485
	EN	<u>7.155</u>	7.078	7.147	7.129	7.230

4.5. Ablation studies

In this section, extensive ablation experiments are conducted to quantitatively and qualitatively validate the rationality of our proposed method and some important hyper-parameters.

4.5.1. Effect of initializing unique features

The initial inputs $\hat{X}$ and $\hat{Y}$ of two UEF play an important role in learning precise unique features. In the proposed LRSP-Fusion, the initial unique features are obtained by subtracting a common feature map from two input source images, where the common feature map is generated by compressing the common feature maps using a 1×1 convolution layer. To verify the impact of initializing unique features on model performance, we conduct an ablation study (denoted as Exp1) by directly replacing the initial unique features in the model with two input images. The results in Table 5 show that our method achieved superior performance on all three datasets. Similarly, the visualization results in Fig. 8

also validate the effectiveness of our proposed approach, for example, our results contain more texture information.

4.5.2. Sparse constraint

To validate the effectiveness of incorporating sparse constraints into the unique features, we replace each stage of UFE (i.e., kernel transposition convolution block (KTCB)) with a residual block (RB), denoted as Exp2. As shown in Table 5, the results demonstrate that enforcing sparsity in the unique feature space effectively enhances the model's performance. From the subjective results in the fourth column of Fig. 8, it can be observed that after removing the sparsity constraint, the fusion results tend to favor one modality, leading to incomplete information representation and noticeable detail loss. To further demonstrate that UFE using Sparse constraint can learn precise unique features, we visualized the features learned by UFE and its variants in MEF, as shown in Fig. 9. In particular, the unique feature maps learned by our UFE more clearly highlight local modality-dependent deviations, such as under-exposed

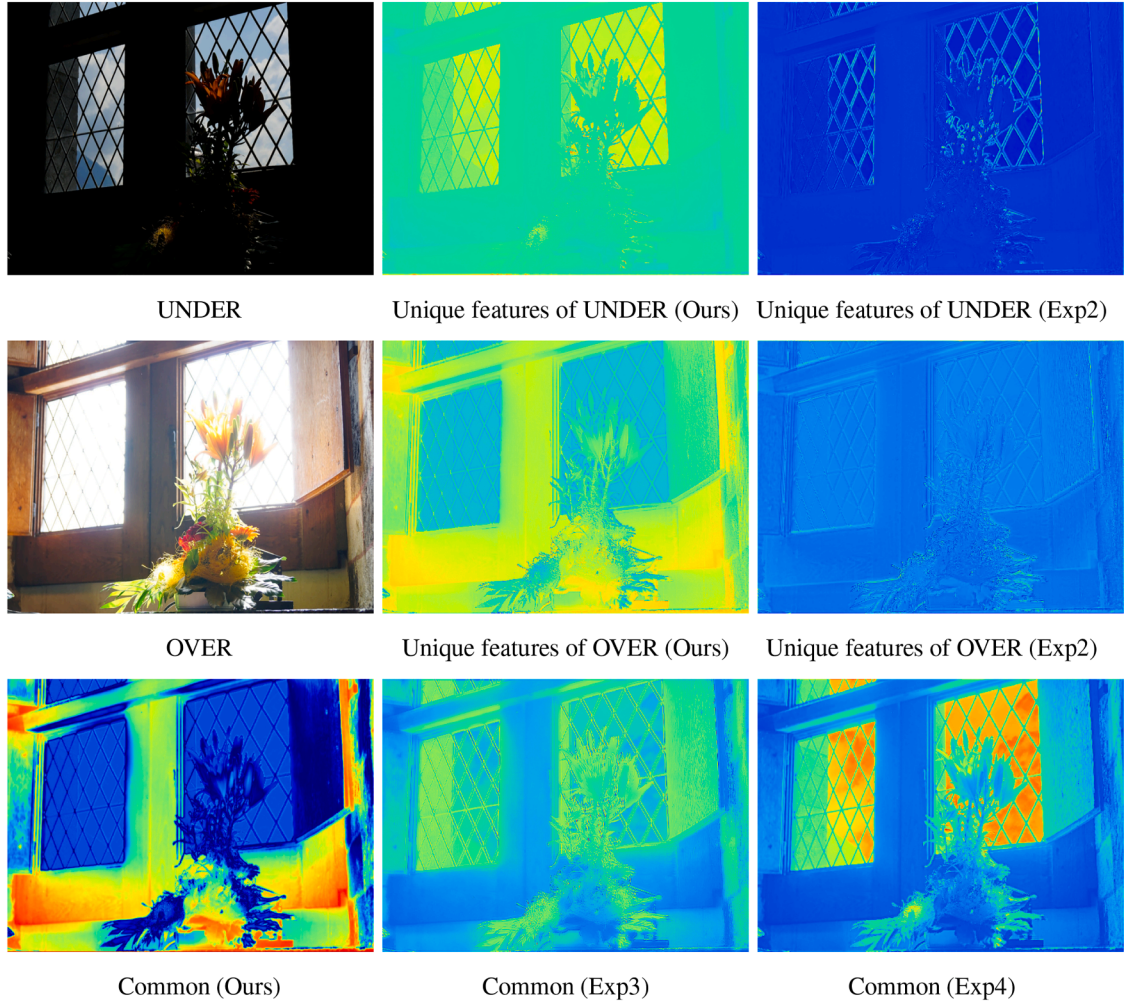


Fig. 9. Visualization of common and unique features.

Table 6
Effectiveness of iteration S in CFE.

	S	3	4	5	6	7	8
IVF	NMI	0.561	0.551	0.554	0.587	0.518	0.551
	SD	41.416	39.715	43.891	48.026	37.048	<u>47.759</u>
	MI	3.783	3.726	3.739	3.948	3.509	3.702
	VIF	0.817	0.8	0.831	0.861	0.755	<u>0.844</u>
	EN	7.017	6.941	7.093	<u>7.169</u>	6.834	7.183
MEF	NMI	0.653	0.738	<u>0.867</u>	1.028	0.854	0.776
	SD	46.123	44.067	<u>59.489</u>	54.52	61.126	41.424
	MI	3.965	4.444	<u>5.256</u>	6.223	5.188	4.701
	VIF	0.782	0.943	<u>1.135</u>	1.23	1.085	0.891
	EN	6.967	6.911	<u>7.146</u>	7.233	6.724	6.739
MFF	NMI	1.043	1.028	1.032	1.048	1.032	1.033
	SD	<u>55.964</u>	53.894	52.586	56.227	53.25	51.734
	MI	<u>7.331</u>	7.234	7.254	7.365	7.259	7.263
	VIF	<u>1.476</u>	1.439	1.455	1.485	1.442	1.382
	EN	<u>7.201</u>	7.092	7.154	7.23	7.146	7.04

Table 7
Effectiveness of iteration K in UFE.

	K	1.0	2.0	3.0	4.0	5.0
IVF	NMI	0.577	0.551	0.587	<u>0.566</u>	0.478
	SD	43.967	42.632	48.026	<u>47.862</u>	42.851
	MI	3.902	3.714	3.948	3.811	3.229
	VIF	0.871	<u>0.863</u>	0.861	0.843	0.770
	EN	7.029	<u>7.064</u>	7.169	7.016	6.985
MEF	NMI	0.902	1.028	<u>0.912</u>	0.663	0.881
	SD	53.867	<u>54.520</u>	55.081	39.347	46.481
	MI	5.500	6.223	<u>5.527</u>	3.974	5.376
	VIF	1.017	1.230	1.182	0.856	<u>1.191</u>
	EN	6.534	<u>7.233</u>	7.242	6.637	7.004
MFF	NMI	0.966	1.048	1.030	<u>1.041</u>	1.021
	SD	50.601	56.227	55.554	55.120	<u>56.119</u>
	MI	6.807	7.365	7.245	<u>7.320</u>	7.185
	VIF	1.326	1.485	1.437	<u>1.468</u>	1.442
	EN	7.001	7.230	7.155	<u>7.192</u>	7.109

and over-exposed regions, while the variant without sparse constraint tends to retain more redundant responses. This visualization provides intuitive evidence that the sparse prior helps preserve salient unique details while suppressing residual common information.

4.5.3. Low-rank decomposition

In our work, we define the common features in two modal images as low-rank background consistency information. Specifically, within the

CFE module, after mapping the two modalities into the shallow feature space, the matrix decomposition (MD) operation is designed to extract these common features. To validate the effectiveness of this design, we conduct two ablation studies. First, since self-attention (SA) is a powerful global feature extraction mechanism, we replace the low-rank MD in CFE with a SA module (denoted as Exp3) modified based on Restormer (Zamir et al., 2022). In addition, we substitute MD with a RB (He et al., 2016) (denoted as Exp4), representing a purely CNN-based architecture.

Table 8
Comparison of params, flops, and average runtime.

Dataset	Metric	U2Fusion	DeFusion	SwinFusion	LRRNet	CDDFuse	GIFNet	Ours
IVF	Size(M)	21.61	7.87	0.93	0.049	1.19	0.61	<u>0.0598</u>
	FLOPs(G)	88.25	15.26	63.73	3.02	166.85	39.81	<u>3.66</u>
	Time(s)	0.142	0.104	1.211	0.019	0.028	<u>0.023</u>	<u>0.047</u>
	Venue	2020, TPAMI	2022, ECCV	2022, JAS	2023, TPAMI	2023, CVPR	2025, CVPR	-
MEF	Method	U2Fusion	DeFusion	SwinFusion	TUFusion	HoLoCo	GIFNet	Ours
	Size(M)	21.61	7.87	0.93	19.07	17.39	<u>0.61</u>	0.0587
	FLOPs(G)	260.7	<u>44.73</u>	187.64	944.25	248.87	115.9	10.66
	Time(s)	0.218	0.115	1.944	<u>0.04</u>	0.059	0.032	0.048
MFF	Venue	2020, TPAMI	2022, ECCV	2022, JAS	2023, TCSVT	2023, Inf.fus	2025, CVPR	-
	Method	U2Fusion	DeFusion	SwinFusion	TUFusion	YDTR	GIFNet	Ours
	Size(M)	21.61	7.87	0.93	19.07	<u>0.22</u>	0.61	0.0587
	FLOPs(G)	374.03	<u>64.18</u>	270.42	1360.81	84.99	166.22	15.12
MFF	Time(s)	0.148	0.051	1.227	0.039	0.087	0.008	<u>0.03</u>
	Venue	2020, TPAMI	2022, ECCV	2022, JAS	2023, TCSVT	2022, TMM	2025, CVPR	-

As presented in Tables 5 and 8, the results clearly demonstrate the superiority of our proposed method, confirming that the use of matrix decomposition in CFE provides more effective extraction of modality-invariant features compared with either SA or RB alternatives. To further demonstrate that the MD operation can learn the rich common features, we visualize the features learned by the original CFE as well as the CFEs in Exp3 and Exp4 in MEF task, as shown in Fig. 9. As can be observed, the common feature maps produced by the original CFE retain more coherent global structures than those generated by the alternative designs, which is consistent with our interpretation that the low-rank prior promotes a compact representation of the dominant shared structure rather than a simple frequency-domain decomposition. Together with the unique-feature visualization above, these intermediate results provide intuitive support for the proposed common/unique separation mechanism.

4.5.4. Effectiveness of iteration number in MD and UFE.

In our network, two key parameters are the iteration number S in the MD in CFE, and the iteration number K in UFE. To evaluate their impact, we conduct six groups of experiments with $S \in \{3, 4, 5, 6, 7, 8\}$, and five groups of experiments with $K \in \{1, 2, 3, 4, 5\}$. As shown in Table 6, for IVF, when $K = 3$, four evaluation metrics achieve the best performance. For MEF, when $K = 2$, three metrics reach the optimum while two others are second-best. Considering the lightweight design of the network, we set $K = 2$. For MFF, the best results are also obtained at $K = 2$. Regarding the value of S , as shown in Table 7, the overall metrics achieve their optimum when $S = 6$. Therefore, we set $S = 6$ in our experiments.

4.6. Model efficiency

To validate the lightweight nature of our network, we also conducted a comparison experiment on model complexity and efficiency. Table 7 shows the total number of training parameters, FLOPs, and average inference time. Specifically, for the IVF task, FLOPs are calculated with an input size of 256×256 , and the average inference time is measured on 42 pairs from the TNO dataset. For the MEF task, FLOPs are computed using the first image from the MEFB dataset, while the average inference time is obtained on 50 image pairs from MEFB. For the MFF task, FLOPs are measured on the first image from the RealMFF dataset, and the average inference time is evaluated on 50 image pairs from the RealMFF dataset. The results show that our approach achieves the lowest parameter and FLOPs, and the second-fastest inference speed among general image fusion frameworks, which may be attributed to the serial design of our network—where UFE depends on the output of CFE. This advantage mainly comes from three aspects: 1) CFE uses low-rank factorization to model shared information without relying on heavy stacked Transformer blocks; 2) JFEB adopts depthwise separable

convolution to reduce parameter overhead; and 3) UFE is implemented with a small number of unfolded stages rather than a very deep black-box encoder. At the same time, the inference speed is not always the fastest, as the current architecture performs common-feature extraction and unique-feature refinement in a serial manner, introducing stage-wise dependency during forward propagation. In future work, we will continue to improve this design in future work to further improve efficiency. Compared with task-specific methods, our framework is equally competitive. For example, our method is only slightly inferior to LRRNet in the IVF task, but surpasses task-specific methods in the MEF and MFF tasks. By exploiting the proposed interpretable architecture, the designed lightweight model significantly reduces both trainable parameters and computational burden, while maintaining strong fusion performance across multiple tasks.

5. Conclusion

In this paper, we propose a novel approach to unify the multi-source image fusion task based on the inherent characteristics of different modal images, which can effectively meet the requirements of different image fusion tasks. By utilizing the low rank and sparse priors of the image matrix, two encoding stages are designed to extract common and unique features of multi-source images, making the feature decomposition process more interpretable. Additionally, an EFFB is constructed to integrate features for precise feature fusion. Experimental results show that LRSP-Fusion outperforms existing methods in multiple image fusion tasks, achieving higher efficiency with fewer parameters. Despite the improved interpretability in feature decomposition, the design of the fusion module remains heuristic. Future work will focus on addressing this limitation by exploring adaptive integration mechanisms and extending the theoretical interpretability to the entire fusion framework.

CRediT authorship contribution statement

Shuying Huang: Writing – original draft, Supervision, Software, Methodology, Formal analysis, Conceptualization; **Xiao Xia:** Writing – original draft, Validation, Software, Methodology, Investigation; **Yong Yang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization; **Weiguo Wan:** Visualization, Validation, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data will be released on <https://github.com/yeager415/LRSP-Fusion>.

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